Differential analysis

Metadata

Analysis

R-Analysis Version	2.11
File	D:_portable\4.2.0-Portable_with_QP-proteomics-DA_proteins_set.txt
Number of rows	2936
Origin File	
Date	26-11-13 11:04:53

Experimental design

File	repBio	Condition
Intensity_10amol_1	inj1	10amol
Intensity_10amol_2	inj2	10amol
Intensity_10amol_3	inj3	10amol
Intensity_10amol_4	inj4	10amol
Intensity_50amol_1	inj1	50amol
Intensity_50amol_3	inj3	50amol
Intensity_50amol_2	inj2	50amol
Intensity_50amol_4	inj4	50amol
Intensity_100amol_1	inj1	100amol
Intensity_100amol_3	inj3	100amol
Intensity_100amol_2	inj2	100amol
Intensity_100amol_4	inj4	100amol
Intensity_250amol_1	inj1	250amol
Intensity_250amol_2	inj2	250amol
Intensity_250amol_3	inj3	250amol
Intensity_250amol_4	inj4	250amol
Intensity_500amol_1	inj1	500amol
Intensity_500amol_3	inj3	500amol
Intensity_500amol_2	inj2	500amol
Intensity_500amol_4	inj4	500amol
Intensity_1fmol_1	inj1	1fmol
Intensity_1fmol_2	inj2	1fmol

File	repBio	Condition
Intensity_1fmol_3	inj3	1fmol
Intensity_1fmol_4	inj4	1fmol
Intensity_5fmol_1	inj1	5fmol
Intensity_5fmol_3	inj3	5fmol
Intensity_5fmol_2	inj2	5fmol
Intensity_5fmol_4	inj4	5fmol
Intensity_10fmol_1	inj1	10fmol
Intensity_10fmol_2	inj2	10fmol
Intensity_10fmol_3	inj3	10fmol
Intensity_10fmol_4	inj4	10fmol
Intensity_25fmol_1	inj1	25fmol
Intensity_25fmol_2	inj2	25fmol
Intensity_25fmol_3	inj3	25fmol
Intensity_25fmol_4	inj4	25fmol
Intensity_50fmol_1	inj1	50fmol
Intensity_50fmol_2	inj2	50fmol
Intensity_50fmol_3	inj3	50fmol
Intensity_50fmol_4	inj4	50fmol

Parameters

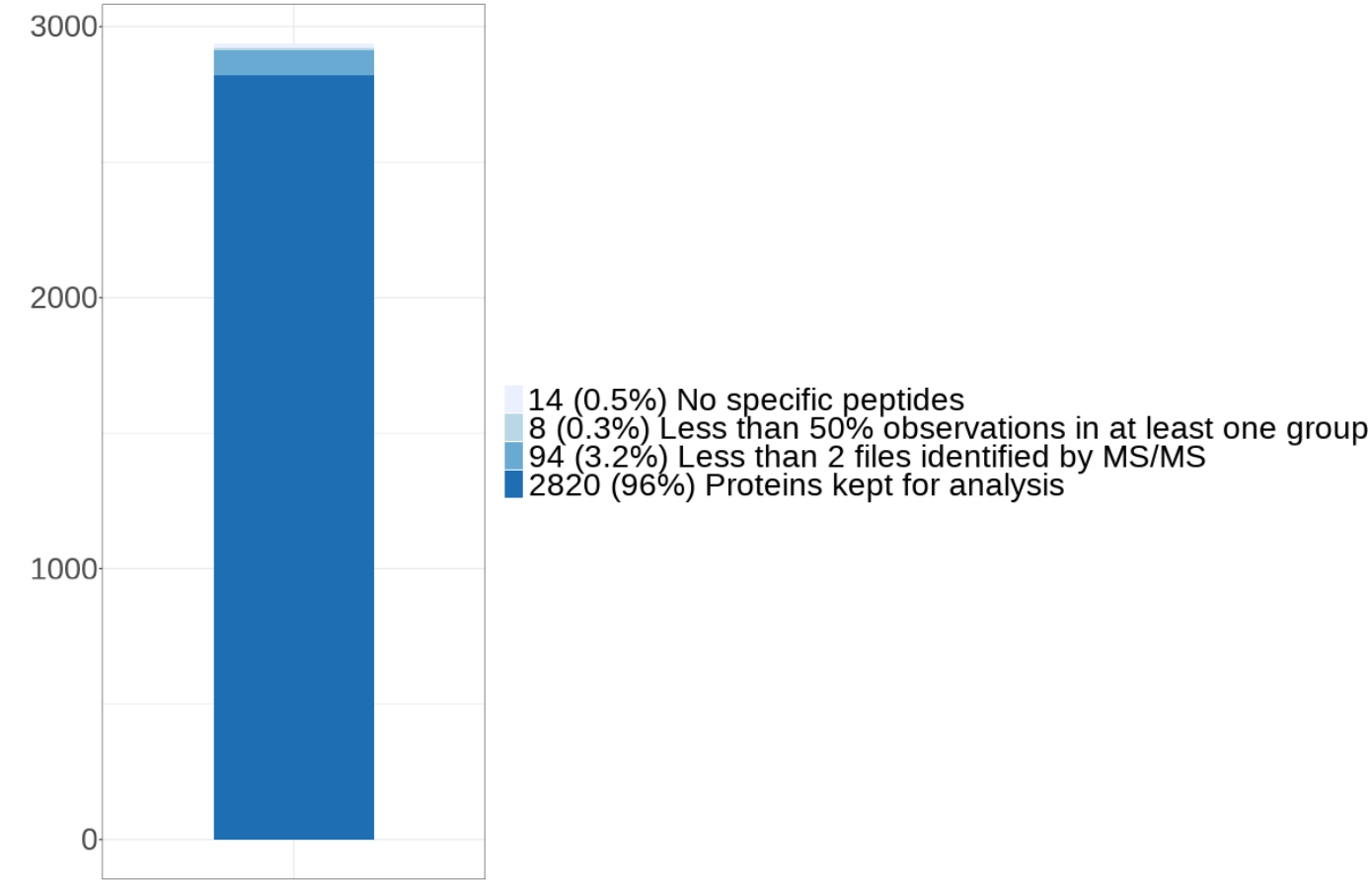
Normalization	T
Proteins.for.normalization	
Filter.threshold.ms	2
Filter.threshold.obs	0.5
No.imputed.values	F
Imputation.MNAR.model	gaussian
Imputation.MNAR.percentile	5
Imputation.MNAR.max.obs	1
Imputation.MCAR.model	none
Imputation.MCAR.min.obs	3
Imputation.MCAR.min.MSMS	3
Imputation.MCAR.intensity.cutoff	auto
Imputation.knn.min.occurrences	3
Test.type	t.test
Test.log	TRUE
Test.alternative	two.sided

Test.paired	FALSE
Test.var.equal	TRUE
Test.adjust.procedure	BH
Stat.threshold.significance	0.05
Stat.threshold.fc	0.58
Volcano.color.style	signifigance
Volcano.manual.accession	
Volcano.no.labeling	F
Comparisons	50fmol/25fmol;50fmol/500amol
Figure.format	PNG
Color.by.group	T

Global Data filtering

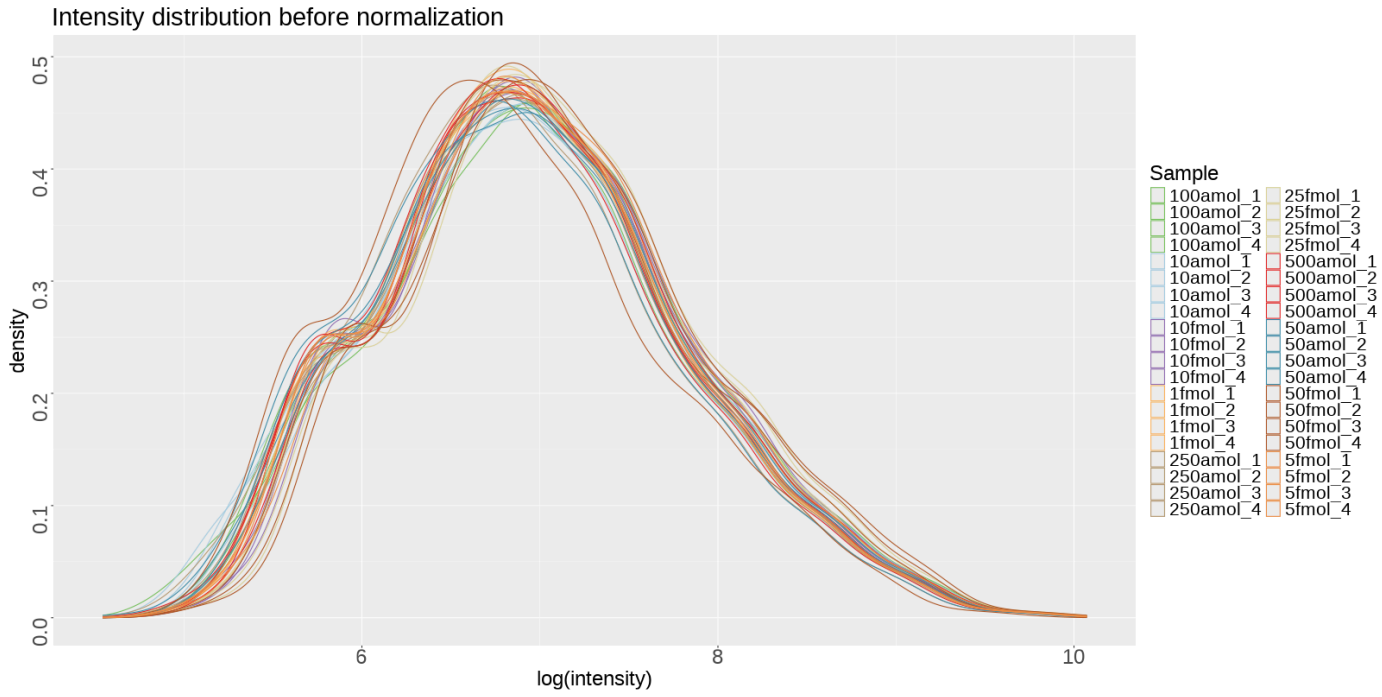
Filtered Proteins

Filter	Value	Value_Percentage
No specific peptides	14	0.5
Less than 50% observations in at least one group	8	0.3
Less than 2 files identified by MS/MS	94	3.2
Proteins kept for analysis	2820	96.0

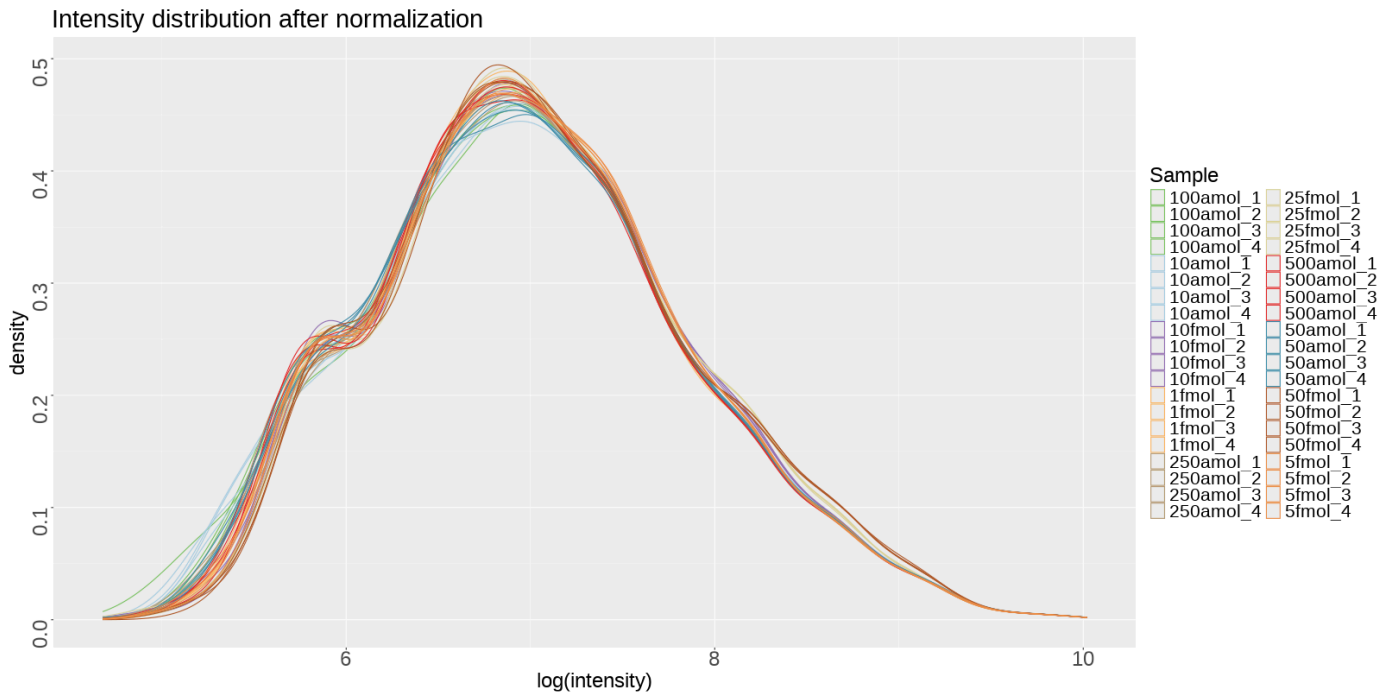


Save figure

Normalization



Save figure



Save figure

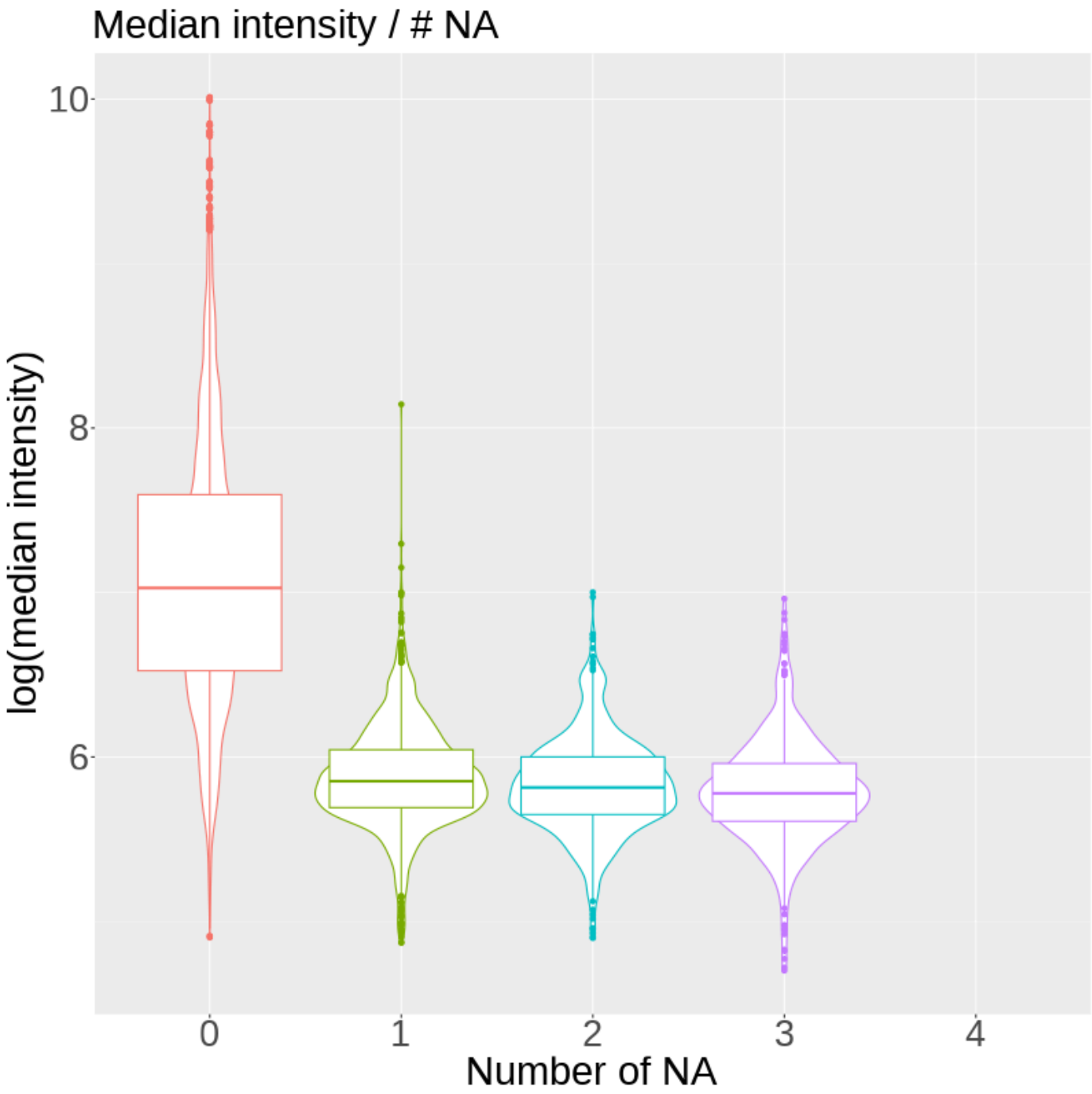
Normalization method : each sample is divided by the median of its ratios to a reference sample (equivalently, the median log2 ratio is subtracted on the log2 scale). The reference is the sample with the lowest number of missing values (Intensity_1fmol_3). The medians were computed on 2820 proteins kept for the analysis. Normalization factors range from 0.579 to 1.141.

Missing values imputation

Distribution of NA values across groups

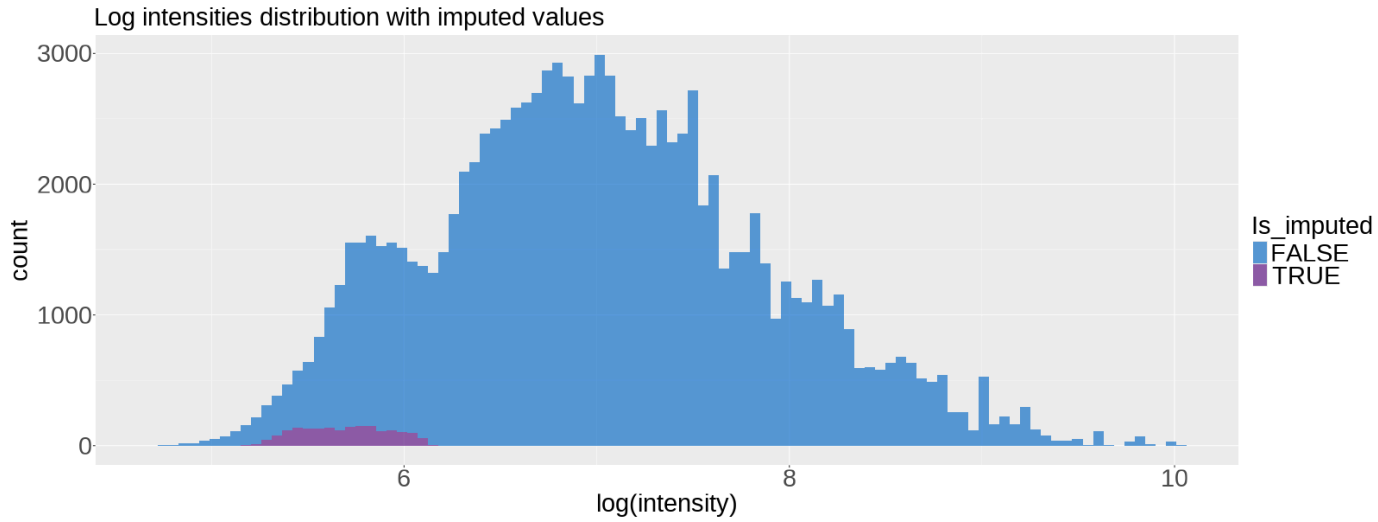
Group	NA_percentage	na_count
10amol	7.271798	854
50amol	6.888624	809
100amol	6.590599	774
250amol	6.692779	786
500amol	5.960490	700
1fmol	5.994550	704
5fmol	5.951975	699
10fmol	5.977520	702
25fmol	6.854564	805
50fmol	7.322888	860
All groups	6.550579	7693

Distribution of NA values across median intensities



Save figure

Imputed values distribution



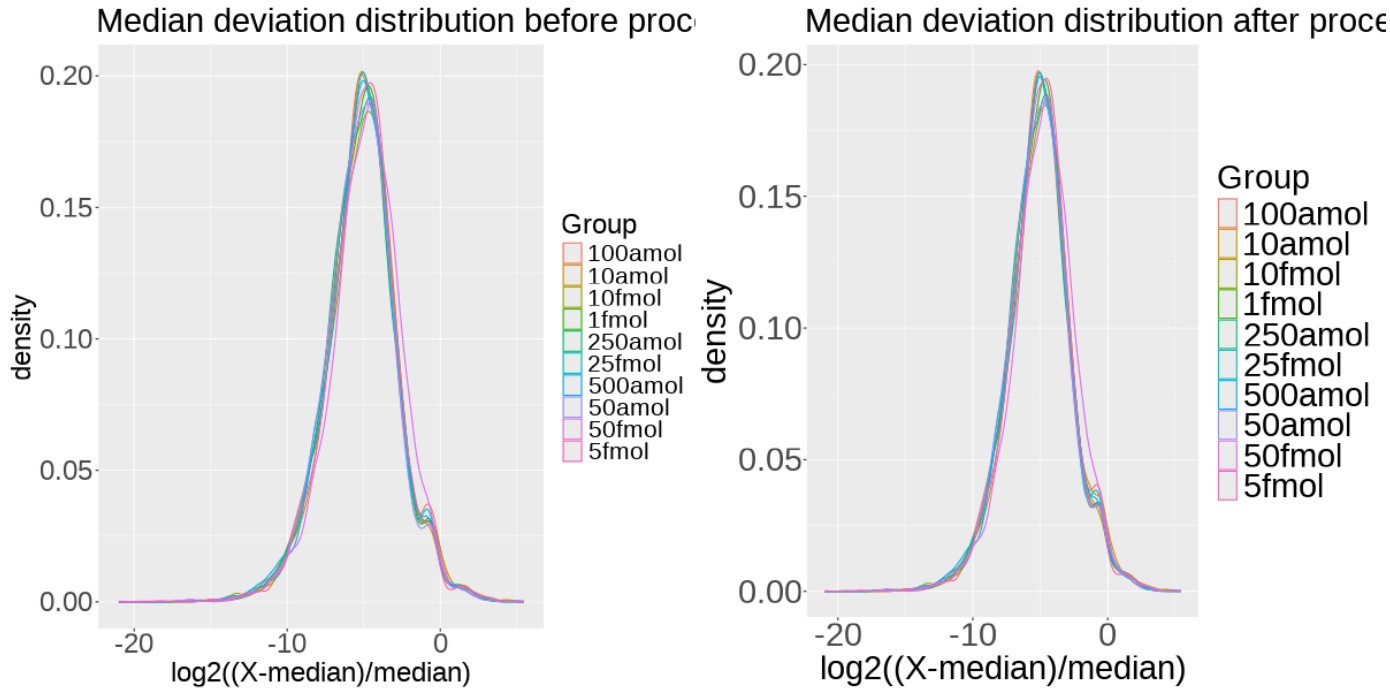
Save figure

Missing values of the 2820 proteins kept for the analysis were imputed with the noise model. The noise level of each sample is the 5% percentile of its observed intensities (log10: 5.57 to 5.79). For each protein and condition with missing values, values are drawn around these levels with a standard deviation of 0.82 log2 (0.25 log10), estimated within conditions on 6564 low-intensity protein/condition measurements. The imputation uses a fixed seed and is reproducible.

Missing values of the proteins kept for the analysis: 4606. MNAR (at most 1 observed value(s) in the condition), imputed with the background noise: 1837. Left missing (tests on the observed values only): 2769.

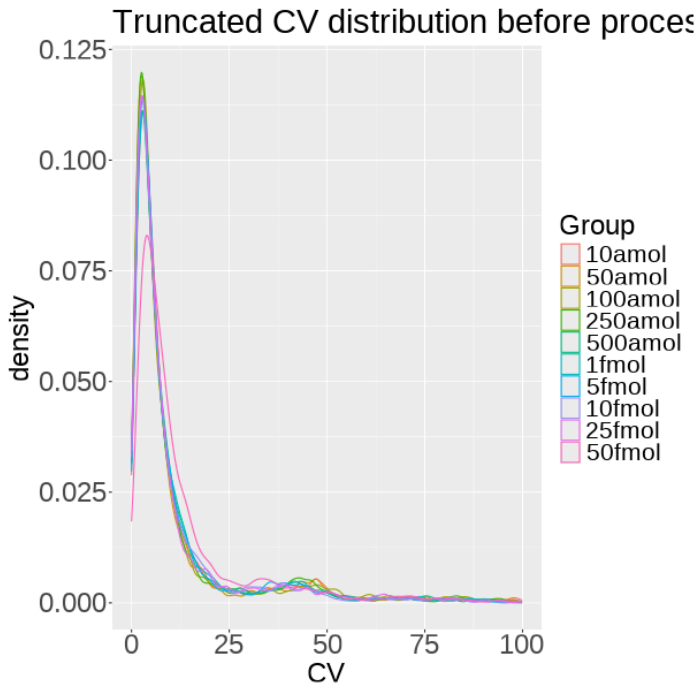
CV distribution

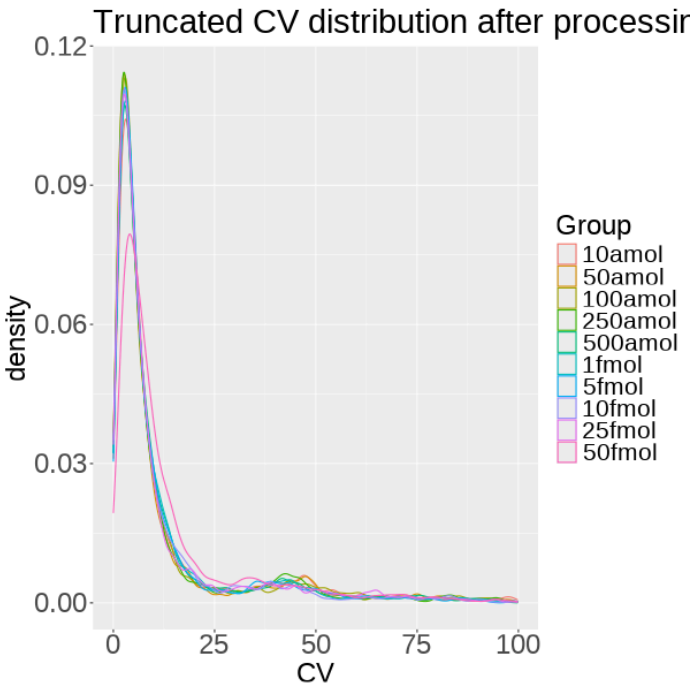
Median deviation



Save figure

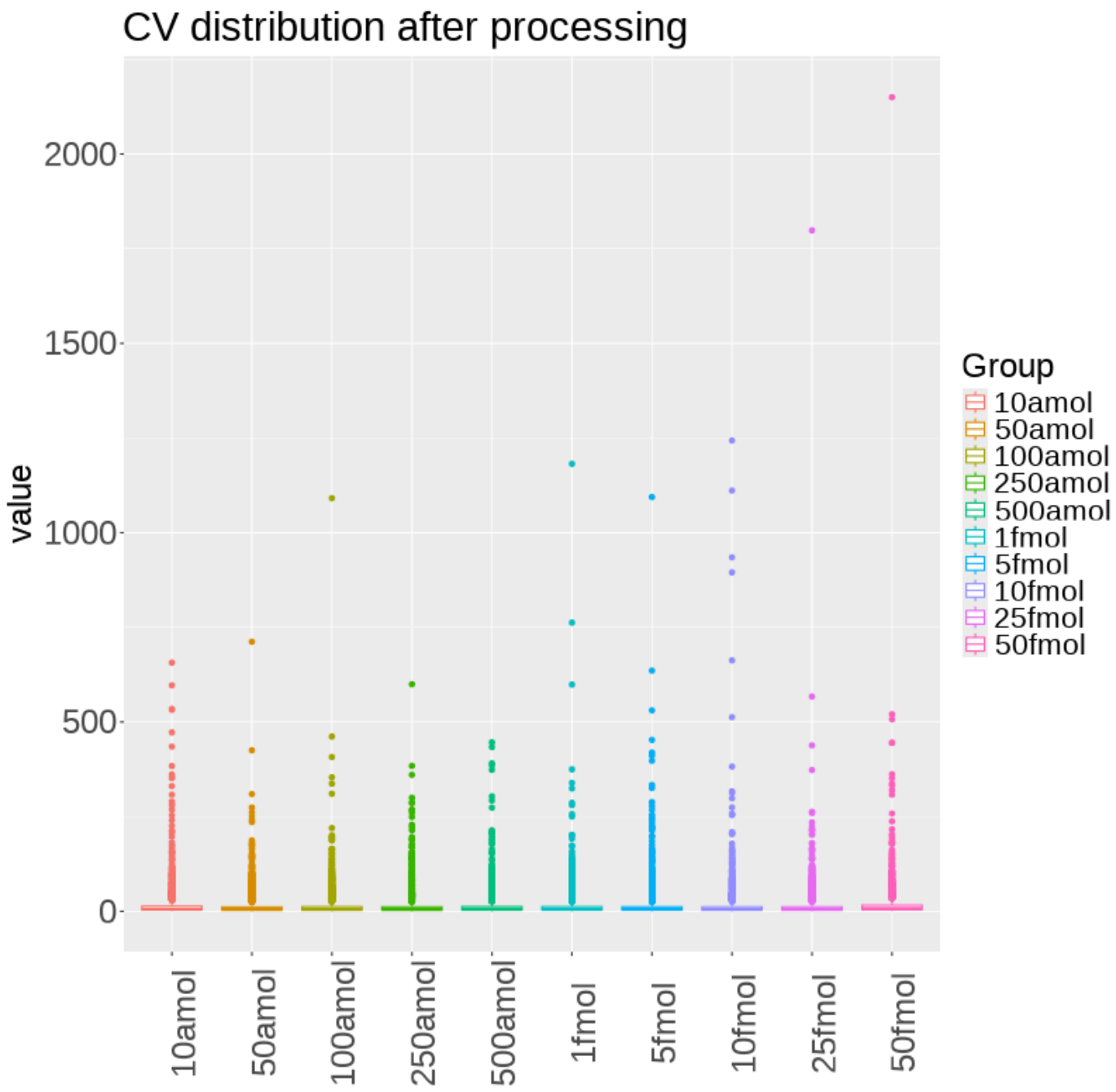
CV distribution



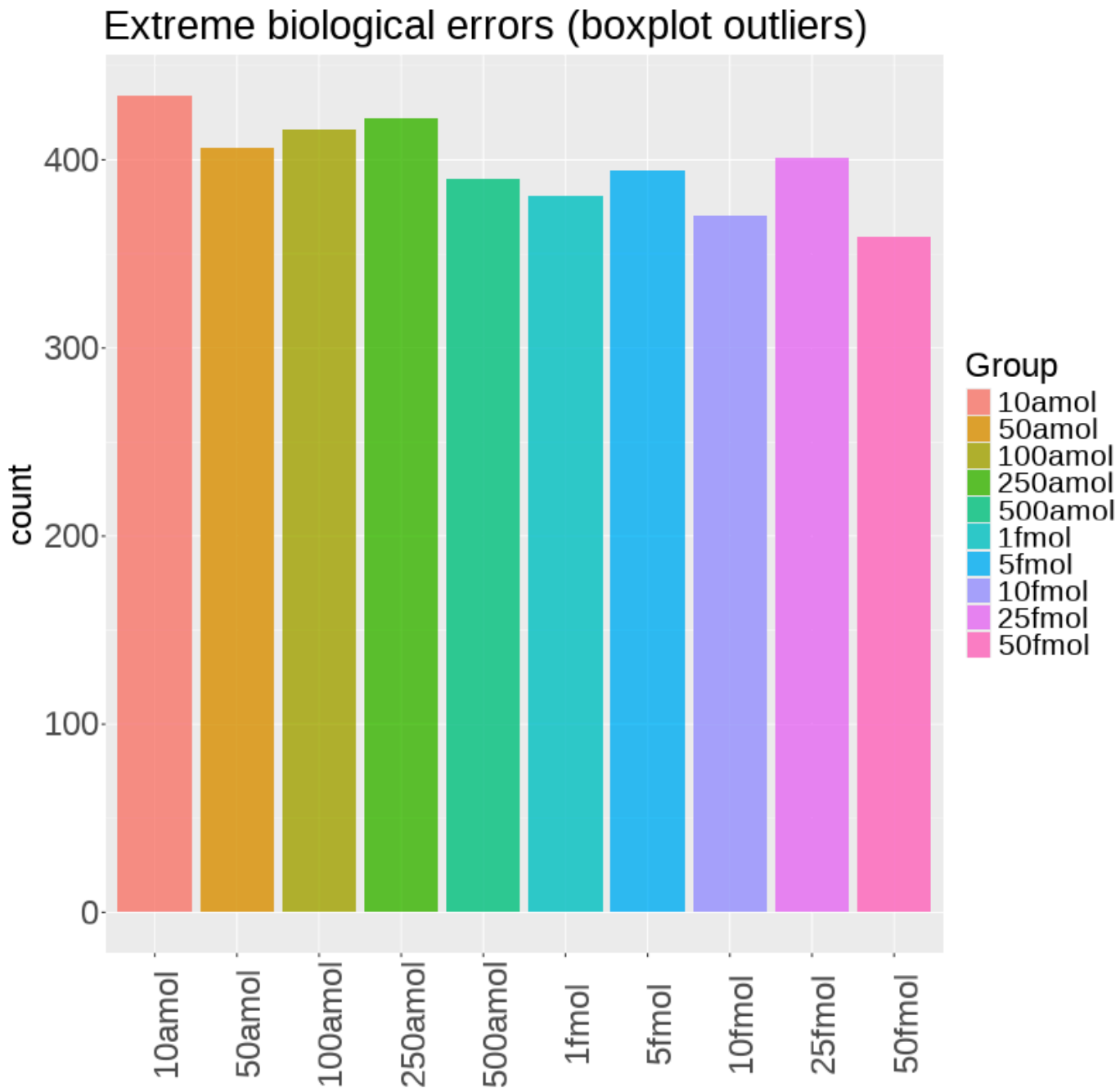


Save figure

Heterogeneity of biological error


CV distribution after processing
value
2000
1500
1000
500
0
10amol
50amol
100amol
250amol
500amol
1fmol
5fmol
10fmol
25fmol
50fmol
Group
10amol
50amol
100amol
250amol
500amol
1fmol
5fmol
10fmol
25fmol
50fmol

Save figure



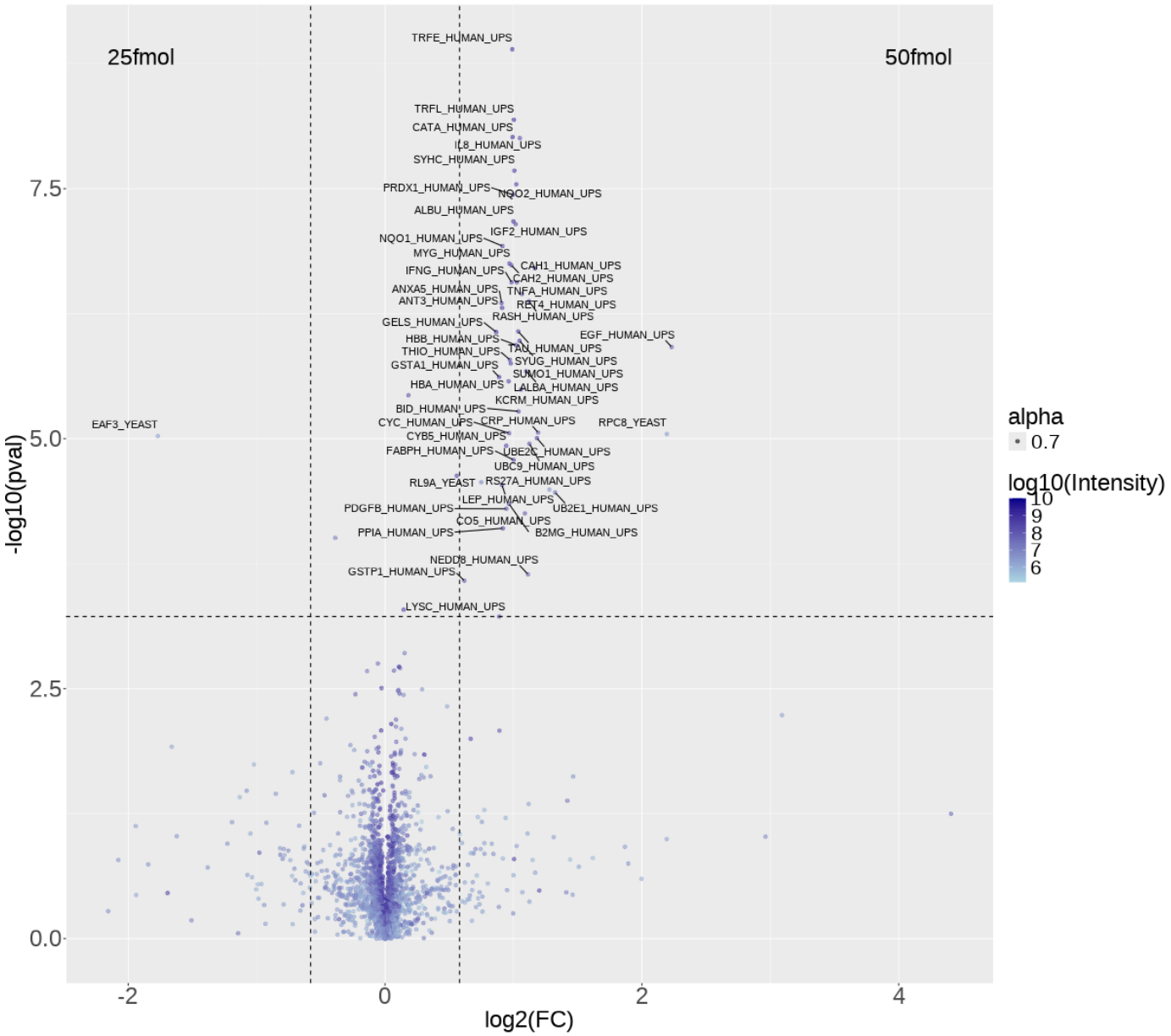
Save figure

Differential analysis

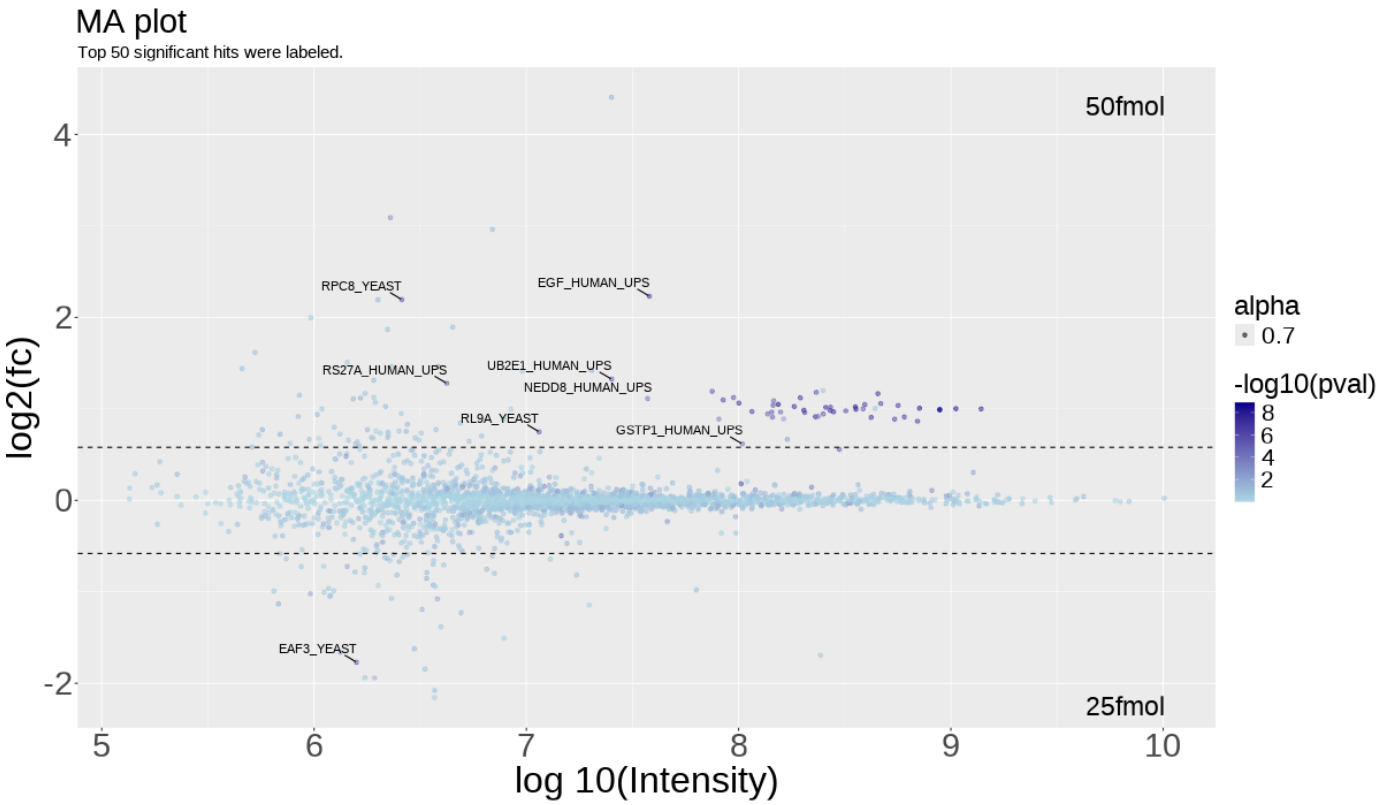
50fmol/25fmol

Filtered Proteins

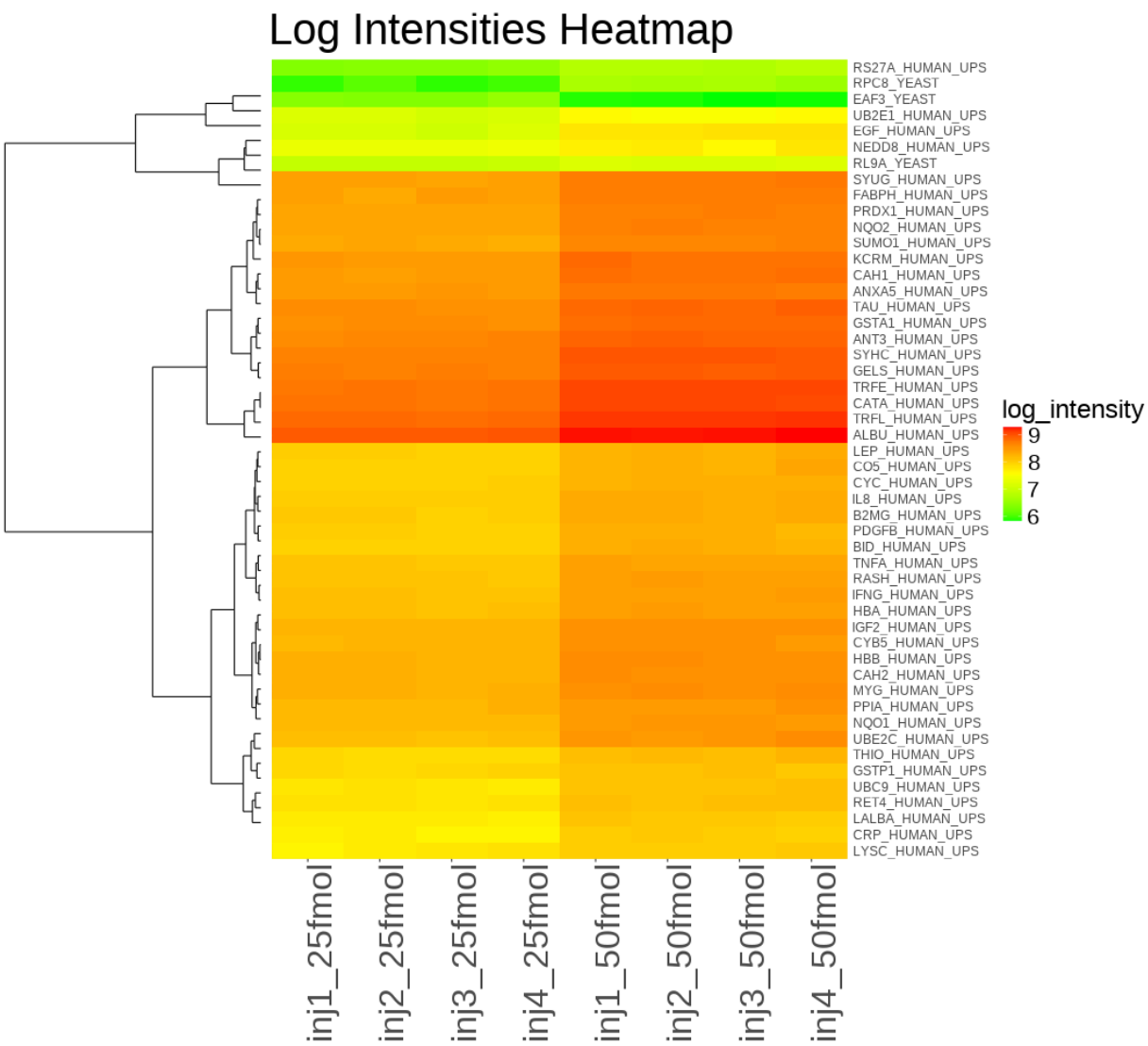
Filter	Value	Value_Percentage
No specific peptides	14	0.5
Less than 50% observations in at least one group	10	0.3
Less than 2 files identified by MS/MS	379	12.9
Proteins kept for analysis	2533	86.3


25fmol
50fmol
TRFE_HUMAN_UPS
TRFL_HUMAN_UPS
CATA_HUMAN_UPS
IL8_HUMAN_UPS
SYHC_HUMAN_UPS
PRDX1_HUMAN_UPS
NQO2_HUMAN_UPS
ALBU_HUMAN_UPS
IGF2_HUMAN_UPS
NQO1_HUMAN_UPS
MYG_HUMAN_UPS
IFNG_HUMAN_UPS
CAH1_HUMAN_UPS
CAH2_HUMAN_UPS
ANXA5_HUMAN_UPS
TNFA_HUMAN_UPS
ANT3_HUMAN_UPS
RET4_HUMAN_UPS
RASH_HUMAN_UPS
GELS_HUMAN_UPS
EGF_HUMAN_UPS
HBB_HUMAN_UPS
TAU_HUMAN_UPS
THIO_HUMAN_UPS
SYUG_HUMAN_UPS
GSTA1_HUMAN_UPS
SUMO1_HUMAN_UPS
HBA_HUMAN_UPS
LALBA_HUMAN_UPS
KCRM_HUMAN_UPS
BID_HUMAN_UPS
EAF3_YEAST
CYC_HUMAN_UPS
CRP_HUMAN_UPS
RPC8_YEAST
CYB5_HUMAN_UPS
FABPH_HUMAN_UPS
UBE2C_HUMAN_UPS
UBC9_HUMAN_UPS
RL9A_YEAST
RS27A_HUMAN_UPS
LEP_HUMAN_UPS
PDGFB_HUMAN_UPS
UB2E1_HUMAN_UPS
CO5_HUMAN_UPS
PPIA_HUMAN_UPS
B2MG_HUMAN_UPS
NEDD8_HUMAN_UPS
GSTP1_HUMAN_UPS
LYSC_HUMAN_UPS
7.5
5.0
2.5
0.0
-log10(pval)
-2
0
2
4
log2(FC)
alpha
0.7
log10(Intensity)
10
9
8
7
6

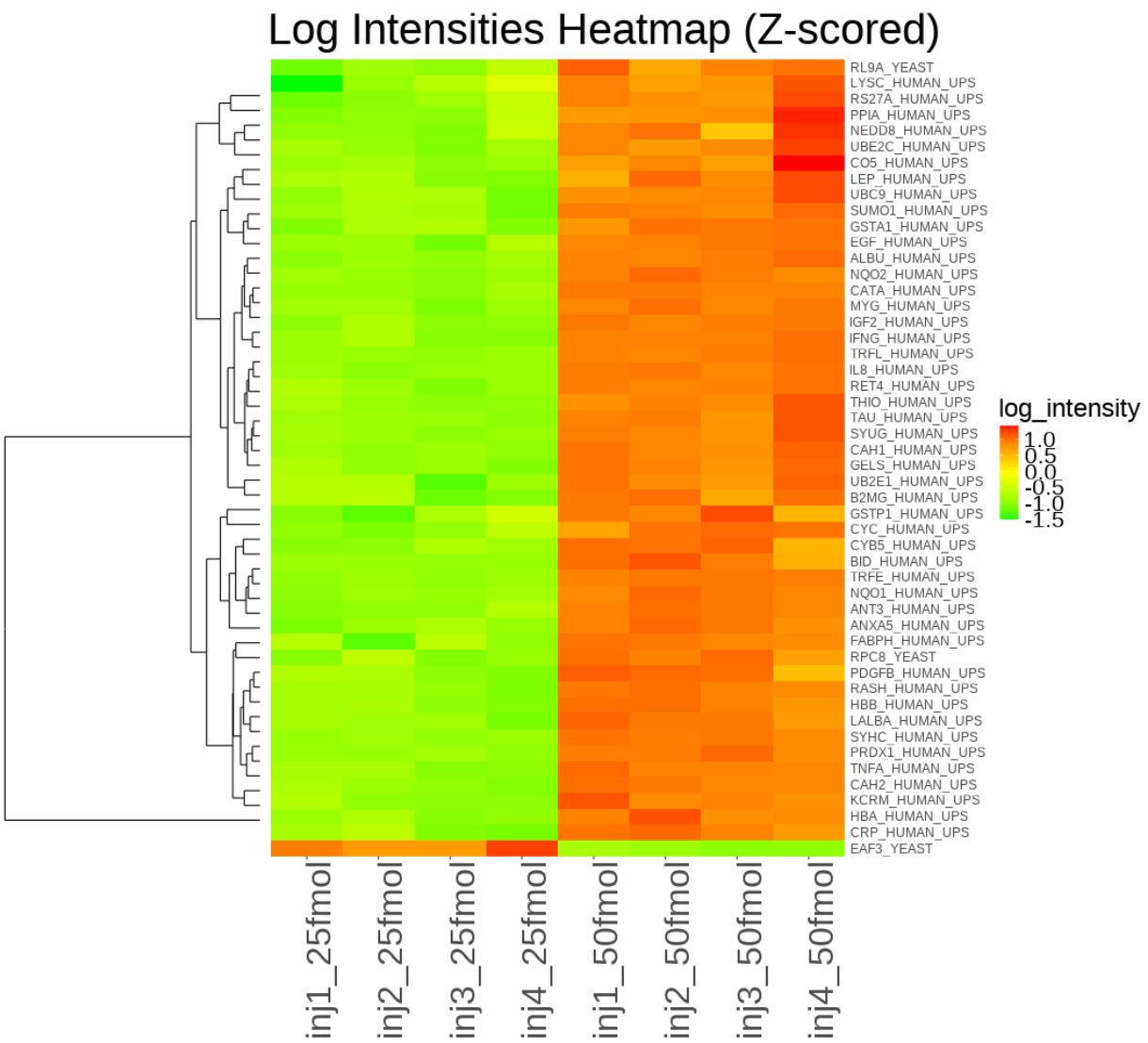
Save figure


MA plot
Top 50 significant hits were labeled.
50fmol
25fmol
RPC8_YEAST
EGF_HUMAN_UPS
RS27A_HUMAN_UPS
UB2E1_HUMAN_UPS
NEDD8_HUMAN_UPS
RL9A_YEAST
GSTP1_HUMAN_UPS
EAF3_YEAST
log2(fc)
log 10(Intensity)
alpha
0.7
-log10(pval)
8
6
4
2

Save figure


Log Intensities Heatmap
RS27A_HUMAN_UPS
RPC8_YEAST
EAF3_YEAST
UB2E1_HUMAN_UPS
EGF_HUMAN_UPS
NEDD8_HUMAN_UPS
RL9A_YEAST
SYUG_HUMAN_UPS
FABPH_HUMAN_UPS
PRDX1_HUMAN_UPS
NQO2_HUMAN_UPS
SUMO1_HUMAN_UPS
KCRM_HUMAN_UPS
CAH1_HUMAN_UPS
ANXA5_HUMAN_UPS
TAU_HUMAN_UPS
GSTA1_HUMAN_UPS
ANT3_HUMAN_UPS
SYHC_HUMAN_UPS
GELS_HUMAN_UPS
TRFE_HUMAN_UPS
CATA_HUMAN_UPS
TRFL_HUMAN_UPS
ALBU_HUMAN_UPS
LEP_HUMAN_UPS
CO5_HUMAN_UPS
CYC_HUMAN_UPS
IL8_HUMAN_UPS
B2MG_HUMAN_UPS
PDGFB_HUMAN_UPS
BID_HUMAN_UPS
TNFA_HUMAN_UPS
RASH_HUMAN_UPS
IFNG_HUMAN_UPS
HBA_HUMAN_UPS
IGF2_HUMAN_UPS
CYB5_HUMAN_UPS
HBB_HUMAN_UPS
CAH2_HUMAN_UPS
MYG_HUMAN_UPS
PPIA_HUMAN_UPS
NQO1_HUMAN_UPS
UBE2C_HUMAN_UPS
THIO_HUMAN_UPS
GSTP1_HUMAN_UPS
UBC9_HUMAN_UPS
RET4_HUMAN_UPS
LALBA_HUMAN_UPS
CRP_HUMAN_UPS
LYSC_HUMAN_UPS
log_intensity
9
8
7
6
inj1_25fmol
inj2_25fmol
inj3_25fmol
inj4_25fmol
inj1_50fmol
inj2_50fmol
inj3_50fmol
inj4_50fmol

Save figure



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Search:

Id	Accession	Label	inj1_25fmol_mean	inj2_25fmol_mean	inj3_25fmol_me
123542	RS27A_HUMAN_UPS	RS27A_HUMAN_UPS	2.14539e+06	2.35042e+06	2.49480e+06
123544	GELS_HUMAN_UPS	GELS_HUMAN_UPS	5.20230e+08	4.84744e+08	4.99236e+08
123637	KCRM_HUMAN_UPS	KCRM_HUMAN_UPS	3.28393e+08	2.96799e+08	2.92895e+08
123660	PDGFB_HUMAN_UPS	PDGFB_HUMAN_UPS	9.94688e+07	9.77180e+07	8.98750e+07
123758	ANT3_HUMAN_UPS	ANT3_HUMAN_UPS	4.01791e+08	4.10430e+08	4.12847e+08
123788	HBB_HUMAN_UPS	HBB_HUMAN_UPS	1.86665e+08	1.87457e+08	1.74125e+08
123876	FABPH_HUMAN_UPS	FABPH_HUMAN_UPS	2.75175e+08	2.21761e+08	2.79956e+08

123891	CAH2_HUMAN_UPS	CAH2_HUMAN_UPS	1.90818e+08	1.83359e+08	1.77998e+08
123916	CRP_HUMAN_UPS	CRP_HUMAN_UPS	4.75732e+07	5.14928e+07	4.23864e+07
123921	SYHC_HUMAN_UPS	SYHC_HUMAN_UPS	4.71443e+08	4.86541e+08	4.62471e+08

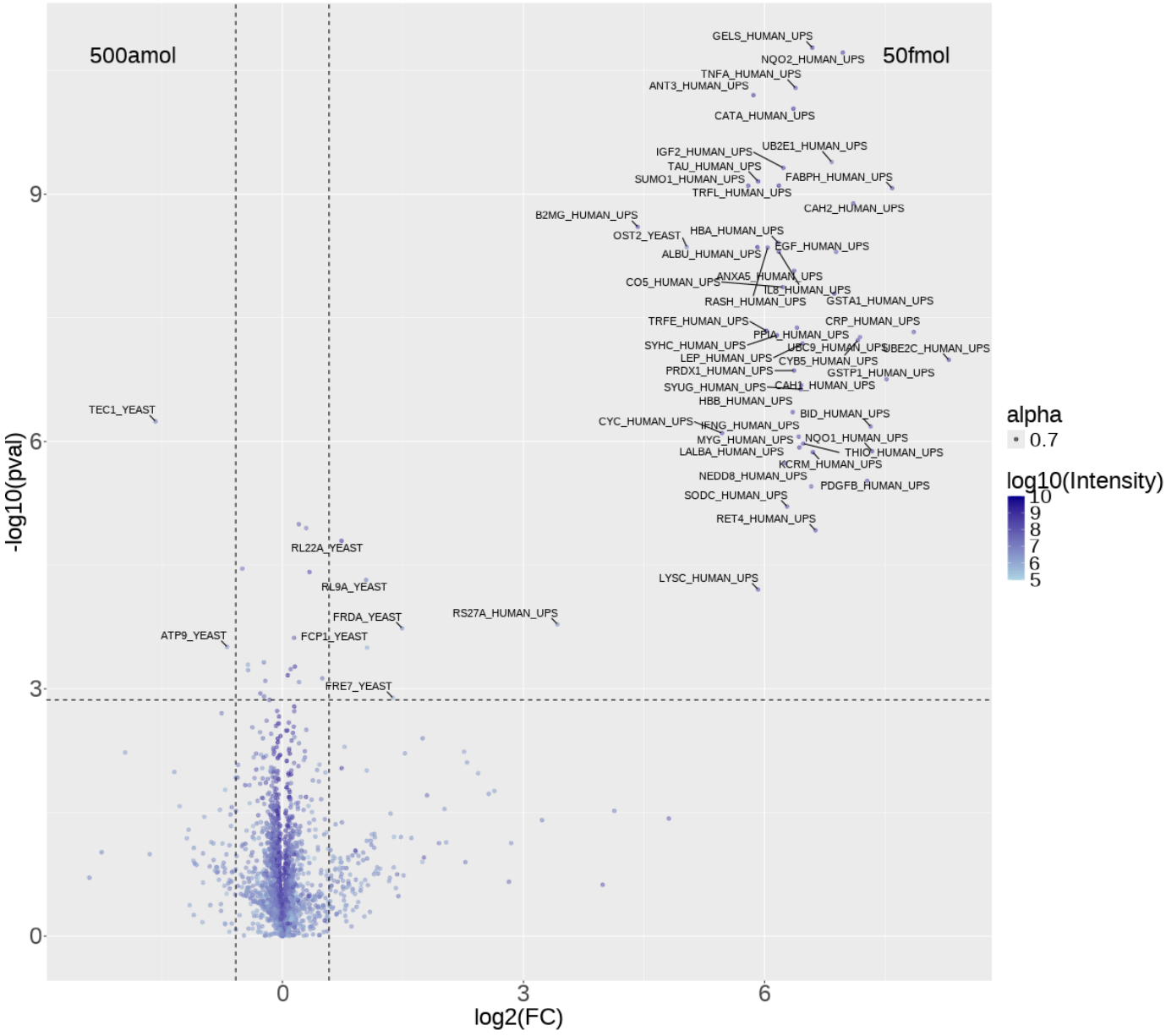
Showing 1 to 10 of 50 entries

Previous 1 2 3 4 5 Next

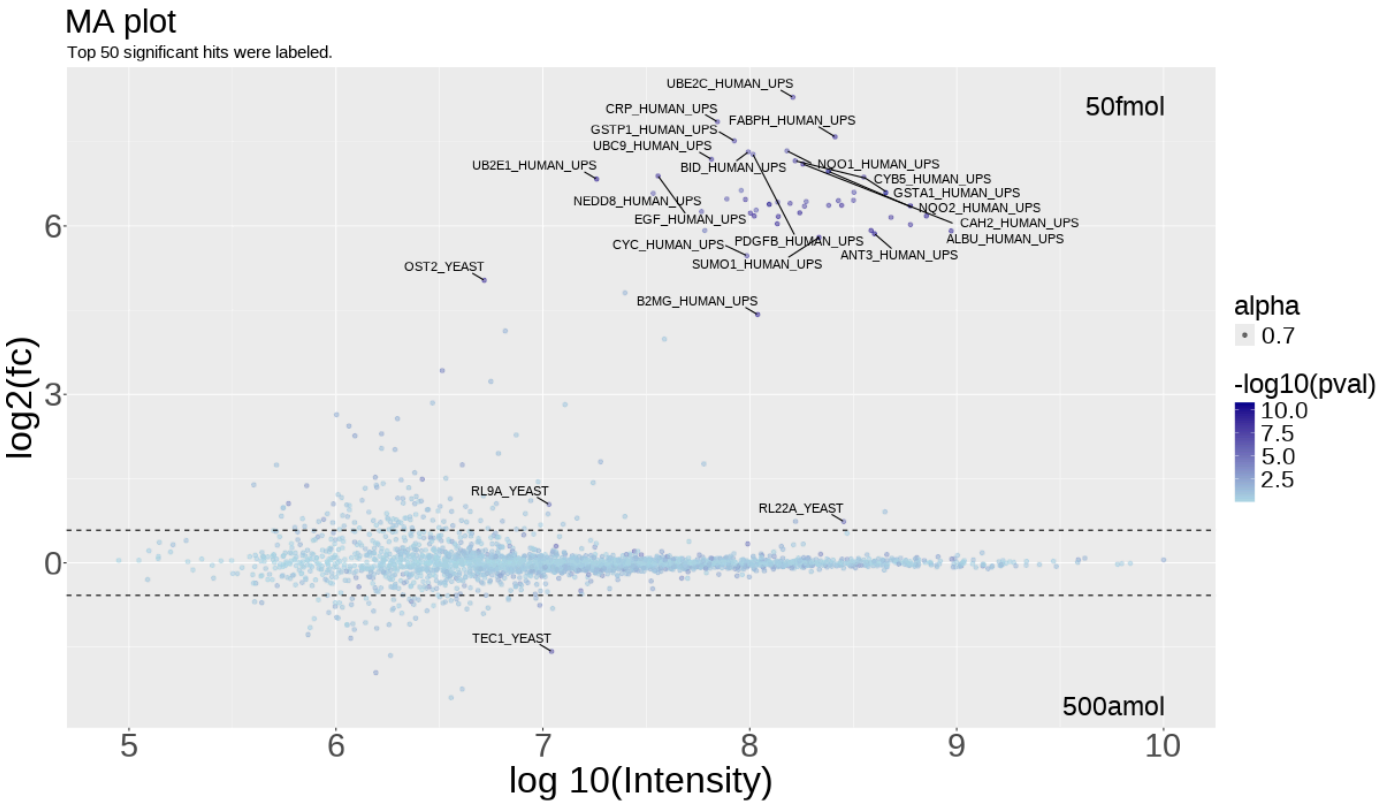
50fmol/500amol

Filtered Proteins

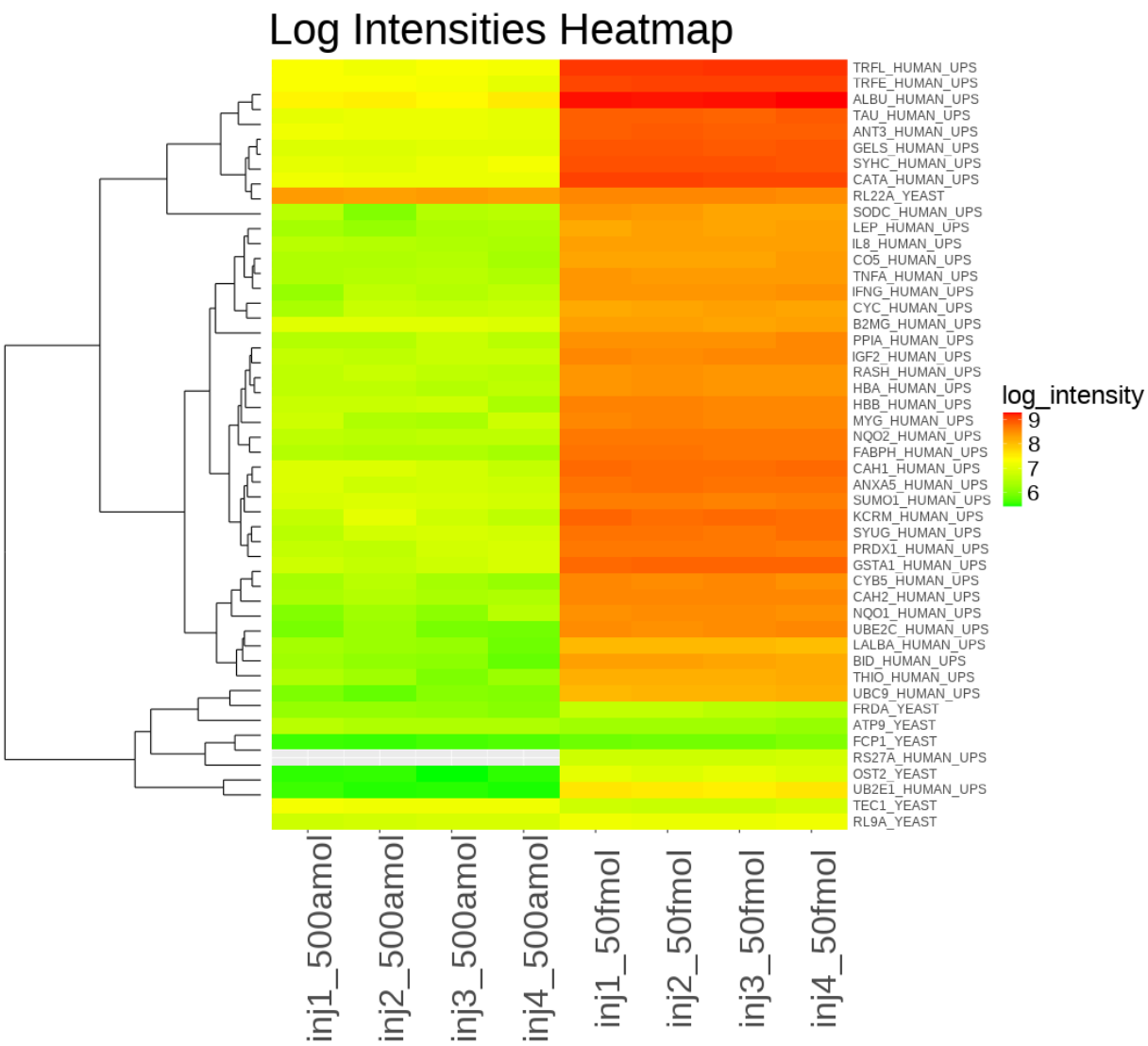
Filter	Value	Value_Percentage
No specific peptides	14	0.5
Less than 50% observations in at least one group	11	0.4
Less than 2 files identified by MS/MS	316	10.8
Proteins kept for analysis	2595	88.4


500amol
50fmol
GELS_HUMAN_UPS
NQO2_HUMAN_UPS
TNFA_HUMAN_UPS
ANT3_HUMAN_UPS
CATA_HUMAN_UPS
IGF2_HUMAN_UPS
UB2E1_HUMAN_UPS
TAU_HUMAN_UPS
SUMO1_HUMAN_UPS
FABPH_HUMAN_UPS
TRFL_HUMAN_UPS
CAH2_HUMAN_UPS
B2MG_HUMAN_UPS
OST2_YEAST
HBA_HUMAN_UPS
EGF_HUMAN_UPS
ALBU_HUMAN_UPS
ANXA5_HUMAN_UPS
CO5_HUMAN_UPS
IL8_HUMAN_UPS
RASH_HUMAN_UPS
GSTA1_HUMAN_UPS
TRFE_HUMAN_UPS
CRP_HUMAN_UPS
PPIA_HUMAN_UPS
SYHC_HUMAN_UPS
UBC9_HUMAN_UPS
UBE2C_HUMAN_UPS
LEP_HUMAN_UPS
CYB5_HUMAN_UPS
PRDX1_HUMAN_UPS
GSTP1_HUMAN_UPS
SYUG_HUMAN_UPS
CAH1_HUMAN_UPS
HBB_HUMAN_UPS
BID_HUMAN_UPS
TEC1_YEAST
CYC_HUMAN_UPS
IFNG_HUMAN_UPS
MYG_HUMAN_UPS
NQO1_HUMAN_UPS
LALBA_HUMAN_UPS
THIO_HUMAN_UPS
KCRM_HUMAN_UPS
NEDD8_HUMAN_UPS
PDGFB_HUMAN_UPS
SODC_HUMAN_UPS
RET4_HUMAN_UPS
RL22A_YEAST
RL9A_YEAST
LYSC_HUMAN_UPS
FRDA_YEAST
RS27A_HUMAN_UPS
ATP9_YEAST
FCP1_YEAST
FRE7_YEAST
-log10(pval)
log2(FC)
alpha
0.7
log10(Intensity)
10
9
8
7
6
5

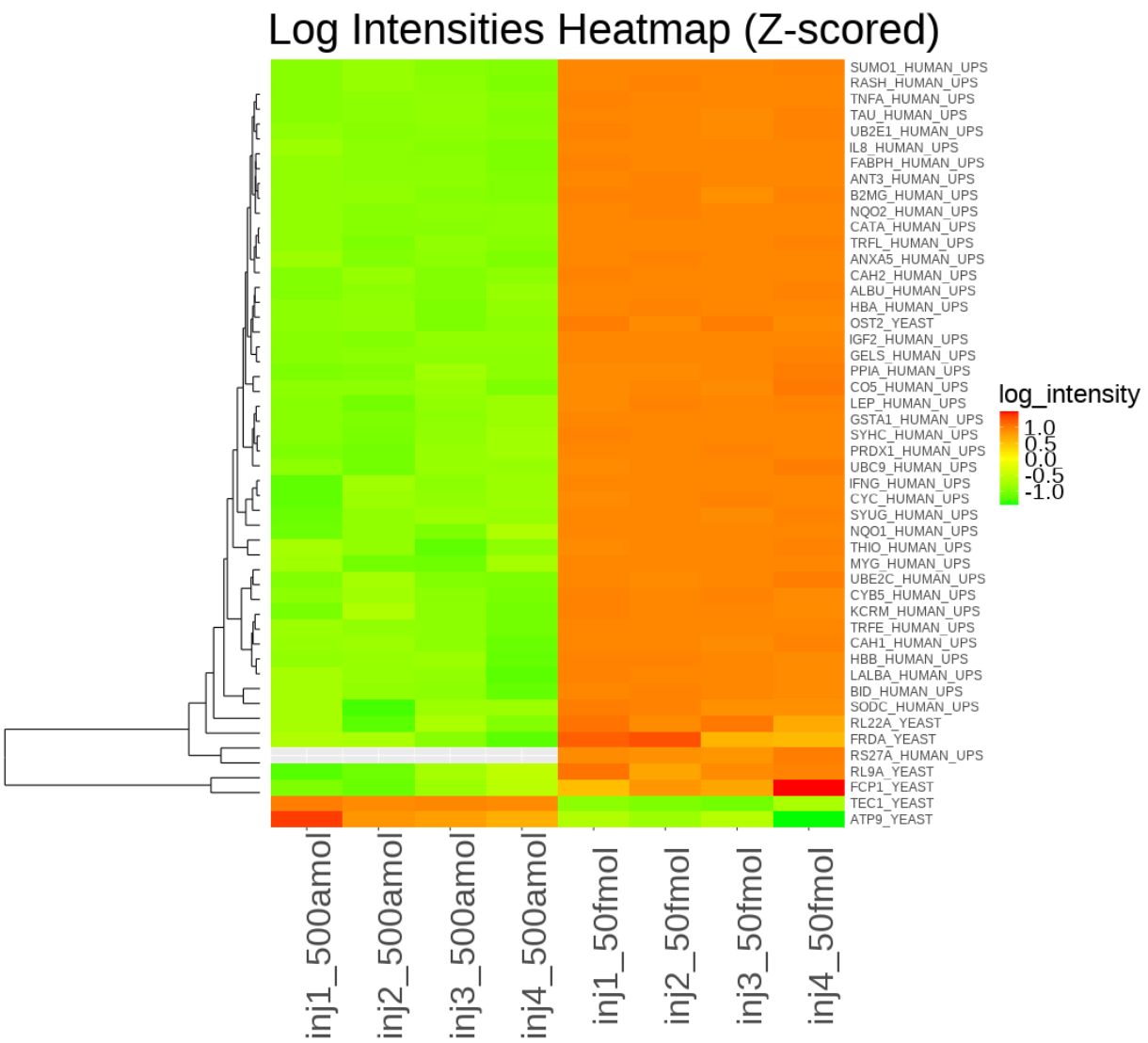
Save figure


MA plot
Top 50 significant hits were labeled.
50fmol
500amol
UBE2C_HUMAN_UPS
CRP_HUMAN_UPS
FABPH_HUMAN_UPS
GSTP1_HUMAN_UPS
UBC9_HUMAN_UPS
UB2E1_HUMAN_UPS
BID_HUMAN_UPS
NQO1_HUMAN_UPS
CYB5_HUMAN_UPS
GSTA1_HUMAN_UPS
NEDD8_HUMAN_UPS
NQO2_HUMAN_UPS
EGF_HUMAN_UPS
CAH2_HUMAN_UPS
ALBU_HUMAN_UPS
CYC_HUMAN_UPS
PDGFB_HUMAN_UPS
ANT3_HUMAN_UPS
SUMO1_HUMAN_UPS
OST2_YEAST
B2MG_HUMAN_UPS
RL9A_YEAST
RL22A_YEAST
TEC1_YEAST
log2(fc)
log 10(Intensity)
alpha
0.7
-log10(pval)
10.0
7.5
5.0
2.5

Save figure


Log Intensities Heatmap
TRFL_HUMAN_UPS
TRFE_HUMAN_UPS
ALBU_HUMAN_UPS
TAU_HUMAN_UPS
ANT3_HUMAN_UPS
GELS_HUMAN_UPS
SYHC_HUMAN_UPS
CATA_HUMAN_UPS
RL22A_YEAST
SODC_HUMAN_UPS
LEP_HUMAN_UPS
IL8_HUMAN_UPS
CO5_HUMAN_UPS
TNFA_HUMAN_UPS
IFNG_HUMAN_UPS
CYC_HUMAN_UPS
B2MG_HUMAN_UPS
PPIA_HUMAN_UPS
IGF2_HUMAN_UPS
RASH_HUMAN_UPS
HBA_HUMAN_UPS
HBB_HUMAN_UPS
MYG_HUMAN_UPS
NQO2_HUMAN_UPS
FABPH_HUMAN_UPS
CAH1_HUMAN_UPS
ANXA5_HUMAN_UPS
SUMO1_HUMAN_UPS
KCRM_HUMAN_UPS
SYUG_HUMAN_UPS
PRDX1_HUMAN_UPS
GSTA1_HUMAN_UPS
CYB5_HUMAN_UPS
CAH2_HUMAN_UPS
NQO1_HUMAN_UPS
UBE2C_HUMAN_UPS
LALBA_HUMAN_UPS
BID_HUMAN_UPS
THIO_HUMAN_UPS
UBC9_HUMAN_UPS
FRDA_YEAST
ATP9_YEAST
FCP1_YEAST
RS27A_HUMAN_UPS
OST2_YEAST
UB2E1_HUMAN_UPS
TEC1_YEAST
RL9A_YEAST
log_intensity
9
8
7
6
inj1_500amol
inj2_500amol
inj3_500amol
inj4_500amol
inj1_50fmol
inj2_50fmol
inj3_50fmol
inj4_50fmol

Save figure



Save figure

Copy CSV Excel PDF Print **Search:**

Id	Accession	Label	inj1_500amol_mean	inj3_500amol_mean	inj2_500amo
123542	RS27A_HUMAN_UPS	RS27A_HUMAN_UPS			
123544	GELS_HUMAN_UPS	GELS_HUMAN_UPS	8.51945e+06	9.63045e+06	9.30008e+06
123637	KCRM_HUMAN_UPS	KCRM_HUMAN_UPS	4.34490e+06	6.07658e+06	1.16070e+07
123758	ANT3_HUMAN_UPS	ANT3_HUMAN_UPS	1.44074e+07	1.36847e+07	1.39867e+07
123788	HBB_HUMAN_UPS	HBB_HUMAN_UPS	4.77230e+06	5.64787e+06	4.97328e+06
123876	FABPH_HUMAN_UPS	FABPH_HUMAN_UPS	3.04214e+06	2.71044e+06	2.82288e+06
123891	CAH2_HUMAN_UPS	CAH2_HUMAN_UPS	2.23890e+06	2.30535e+06	3.14865e+06

123921	SYHC_HUMAN_UPS	SYHC_HUMAN_UPS	1.19088e+07	1.40171e+07	9.66042e+06
123928	RL22A_YEAST	RL22A_YEAST	2.23246e+08	2.25118e+08	1.96120e+08
123956	TEC1_YEAST	TEC1_YEAST	1.71507e+07	1.65403e+07	1.60440e+07